

# Hollow M53: Boss Roster + Boss Framework V1

Engineer-facing handoff for the first 10-boss roster. M53 adds seeded boss selection, fixed-HP boss definitions, unique Room Designer-compatible arenas, top-center boss HUD, and a runtime behavior layer while preserving existing boss rewards.

## Roster Lock

| Band | Boss              | HP | Identity                                       | Arena                 |
|------|-------------------|----|------------------------------------------------|-----------------------|
| W1   | Stone Warden      | 24 | Slow pursuit, charge, stomp burst, 4-way phase | Broken Gateway        |
| W1   | Splinter Saint    | 22 | Side hops and short radial splinters           | Narrow Shrine         |
| W1   | Gravel Maw        | 28 | Fast snake chase, burrow, summon pressure      | Sandy Pit-Ring        |
| W2   | Cartouche Widow   | 32 | Falling-shot spread pressure                   | Open Tomb             |
| W2   | Iron Reliquary    | 36 | Cover shooter, peek shots, relocation          | Cover Maze            |
| W2   | Mirror Husk       | 34 | Splits pressure bodies at HP thresholds        | Symmetric Mirror      |
| W2   | Ash Comet         | 38 | Dash/jump impacts and radial bursts            | Charred Crossing      |
| W3   | Choir of Teeth    | 42 | Pattern bursts capped at 24 projectiles        | Hell/Heaven Dais      |
| W3   | Rust Bishop       | 46 | Beam windups, mine rings, short dashes         | Industrial Cover Grid |
| W3   | Hollow Star Larva | 50 | Chase, summon, starfall, desperation burst     | Blind Deep            |

## Runtime Contract

- Bosses are chosen deterministically from boss catalog, branch seed, world index, and boss room id.
- World 1 selects HP 20-30 bosses, World 2 selects HP 31-40 bosses, and World 3 selects HP 41-50 bosses.
- Boss rooms keep the existing boss reward pool and boss-key lock rules.
- Boss HP is fixed by boss definition; no hidden world stat scaling is applied.
- Boss-owned projectiles are capped at 24 active projectiles.
- Summoned minions use existing enemy kinds and keep room clear gated until boss and remaining summons are dead.

## Implementation Map

| Area                                   | Purpose                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------|
| BossDefinition / BossCatalogDefinition | Roster data: HP, world band, behavior id, arena id, phase and attack metadata.                 |
| BossSelectionResolver                  | Seeded world-banded selection for normal runs and fixed-seed challenges.                       |
| RoomEncounterAssignment metadata       | Persists bossId, bossArenaId, bossWorldBand, and phaseState in encounter plan saves.           |
| BossRuntimeController                  | Bespoke v1 behavior for charge, hop, burrow, split pressure, projectile patterns, and summons. |
| BossHudController                      | Top-center name, HP bar, and compact status line on PlatformShellCanvas.                       |
| Milestone53AssetGenerator              | Creates boss catalog, boss arenas, docs, reports, scene wiring, and curated draft mirroring.   |

## Validation Checklist

- Catalog has exactly 10 bosses and Stone Warden is fixed at 24 HP.
- Every boss has display name, behavior id, world band, arena id, attacks, and fixed HP 20-50.
- Seeded boss selection is deterministic and respects world band.
- Arena runtime JSON imports through HollowRuntime V2 and remains Room Designer-compatible.
- Boss bar appears only while a live boss is active and stays on PlatformShellCanvas.
- Boss projectiles use boss cap; summons count through existing enemy registration.
- Existing boss rewards, boss-key locks, hub/world loop, challenge seeds, minimap, saves, chests, and platform scaling remain stable.

## Rafal ArtPass Targets

M53 still uses generated placeholders. Rafal can replace the shared active EnemyBoss ArtPass target first, then create optional silhouette variants named after boss IDs: AP\_Boss\_StoneWarden, AP\_Boss\_SplinterSaint, AP\_Boss\_GravelMaw, AP\_Boss\_CartoucheWidow, AP\_Boss\_IronReliquary, AP\_Boss\_MirrorHusk, AP\_Boss\_AshComet, AP\_Boss\_ChoirOfTeeth, AP\_Boss\_RustBishop, AP\_Boss\_HollowStarLarva. All remain visual-only children with no gameplay colliders or scripts.